

UNIVERSITY OF RWANDA
COLLEGE OF SCIENCE AND TECHNOLOGY
SCHOOL OF ENGINEERING
DEPARTMENT: CIV-WE AND STRUCTURE/Year 1, CATH/15 MARKS
MAT 1166: MATHEMATICS FOR ENGINEERS I.

DATE: 28/06/2021

DURATION: 1H : 30'

QUESTION 1:(3 Marks)

✓ Find the area of the region enclosed by the parabola $y = x^2$ and $y = 2x - x^2$.

✓ **QUESTION 2:(2 Marks)**

The first term of an arithmetic progression is 25 and 5th term is 20

(a) What is the reason of the progression?

(b) What is the 10th term?

✓ **QUESTION 3:(3 Marks)**

Find the interval and radius of convergence of the series $\sum_{n=1}^{+\infty} \frac{(x-5)^n}{n^2}$.

✓ **QUESTION 4:(3 Marks)**

Find the taylor polynomial of degree 5 for $\sin(x)$ about $x = \pi$.

✓ **QUESTION 5:(4 Marks)**

Solve the following system
$$\begin{cases} x_1 + x_2 = 5 \\ 2x_1 + x_2 - x_3 = 6 \\ 3x_1 - x_2 + 2x_3 = 7 \end{cases}$$

COLLEGE EXAMINATIONS – ACADEMIC YEAR 2015- 2016

SCHOOL OF SCIENCE

DEPARTMENT OF APPLIED MATHEMATICS

FIRST YEAR SEMESTER I

FINAL EXAMINATION

MATHEMATICS FOR ENGINEERERS I (MAT1166)

DATE: .../December/2015

TIME: 2 hours

MAXIMUM MARKS = 50

INSTRUCTIONS:

1. This paper contains **TWO** sections.
 2. Section A is compulsory, and Answer any **TWO** of the **THREE** questions in Section B.
 3. No written materials allowed into the Examination Room.
 4. Write all your answers in the answer booklet provided.
 5. Do not forget to write your Registration Number.
 6. Do not write any answers on this question paper.
 7. Where appropriate draw large clearly labeled diagrams in your answers.
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SECTION A / 20 Marks**QUESTION 1.**

A. Find the Taylor approximation of $f(x) = \sin x$ about $x = \pi$ (3 Marks)

B. Form an arithmetic progression which has sum equal to 70 and, the first term equal to -29 and $n = 7$.

C. Find $\frac{dy}{dx}$ if $x^3 + y^3 = 6xy$ (2 Marks)

D. Evaluate $\int_{-\infty}^{+\infty} \frac{dx}{1+x^2}$ (3 Marks)

E. A box with a square base and an open top must have a volume of 32000cm^3 . Find the dimension of the box that minimizes the amount of materials used. (Hint: Minimize total area). (3 Marks)

F. Find the volume of the solid obtained by rotating the region bounded by $y = x^3$, $y = 8$ and $x = 0$ about y axis. (3 Marks)

G. Show that the vectors $\vec{A} = \langle 1, 4, -7 \rangle$, $\vec{B} = \langle 2, -1, 4 \rangle$ and $\vec{C} = \langle 0, -9, 18 \rangle$ are coplanar in same plane (3 Marks)

SECTION B / 30 Marks**QUESTION 2.**

a) Obtain the reduction formula for $I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx$ (6 Marks)

b) State the limit comparison principle and use it to determine the convergence or

divergence of the series $\sum \frac{4\sqrt{n}-1}{n^2+2\sqrt{n}}$ (5 Marks)

c) From the orthogonal set $\{(1, 1, 1)^T, (2, 1, -3)^T, (4, -5, 1)^T\}$ form orthonormal set in $\mathbb{R}^3$ (4 Marks)

QUESTION 3.

- a) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined as $T(\vec{x}) = \langle x_2, x_1, x_1 + x_2 \rangle$. Show that T is a linear operator? (4 Marks)
- b) Calculate the area of surface S ranging between the parabola $P \equiv y = x^2 + 1$ and the line $D \equiv x + y = 3$ on the interval $[0, 2]$. (5 Marks)
- c) Find the radius and interval of convergence of series:

$$\sum_{n=0}^{\infty} \frac{(-3)^n x^n}{\sqrt{n+1}} \quad (6\text{Marks})$$

QUESTION 4.

- a) Let $A = \begin{bmatrix} 2 & -3 & 1 \\ 1 & -2 & 1 \\ 1 & -3 & 2 \end{bmatrix}$. Find the Eigen-values and the corresponding Eigen-vectors of a matrix A ? (8Marks)
- b) Determine where the function $f(x) = 4x^3 - 3x^2 - 6x + 12$ is increasing, decreasing, concave up, concave down, inflection points, maximum and minimum points. (7Marks)